Exercise 7.3

1. Work out each of these without using a calculator:

(i) Increase 90 by 10% (ii) Increase 60 by 20% (iii) Increase 150 by 30%

(iv) Decrease 40 by 10% (v) Decrease 120 by 20% (vi) Decrease 200 by 5%

2. Work out these without using a calculator:

Increase

(i) 100 by 60% (ii) 300 by 65% (iii) 320 by 45%

(iv) 280 by 15% (v) 480 by 5% (vi) €80 by 8%

3. Write these percentages as decimals:

(i) 6% (ii) 15% (iii) 35% (iv) 70% (v) $7\frac{1}{2}$%.

4. To find 82% of 600, multiply 600 by 0.82 on a calculator.
Now fill in the missing decimal in each of the following:
(i) To find 38% of 700, multiply 700 by ☐
(ii) To find 60% of 580, multiply 580 by ☐
(iii) To find 13% of 270, multiply 270 by ☐
(iv) To find 6% of 300, multiply 300 by ☐
(v) To find 120% of 640, multiply 640 by ☐
(vi) To find $12\frac{1}{2}$% of 800, multiply 800 by ☐

5. (i) Increase 240 by 10%.
(ii) Increase 150 by 40%.
(iii) Decrease €140 by 15%.
(iv) Decrease 240 kg by $12\frac{1}{2}$%.
(v) Increase €90 by 8%.
(vi) Decrease 80 kg by 35%.

6. Find the weight of each of the following after the extra has been added:
(i) A 450 g packet of cornflakes with 30% extra
(ii) A 750 g container of yogurt with 10% extra
(iii) A 250 g pizza with 15% extra.

7. If 11% of a number is 77, find the number.

8. If 20% of a sum of money is €70, find the sum of money.

9. If 15% of the price of a car is €2775, find the price of the car.

10. A bus company plans to increase its prices by 8%. At the moment, the bus fare to town is €1.50. Calculate the new fare.

11. The shop price of a CD is €19.95.
Calculate the online price if you save $22\frac{1}{2}$%.
Give your answer to the nearest cent.

12. Deirdre spent 40% of her money in one shop and 35% of it in another.
If she had €26.80 left, how much had she at first?

13. In a school, there are 336 girls out of a total enrolment of 560 pupils.
Find the percentage of boys in the school.

14. Dara got a wage increase of 5%.
If his new wage is €892.50, find his wage before the increase.

15. A car is valued at €26 000. It depreciates in value by 20% in the first year and in the second year, it depreciates by 15% of its value at the beginning of the year.
Find the value of the car
(i) after 1 year (ii) after 2 years.